Course Code: CBS3005

Course Title: Cloud, Microservices and Applications

Class Number: VL2025260105679

Lab Assessment - 4

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Q. A company wants to deploy a secure, scalable, and highly available web application on AWS for global users. Perform the following tasks in AWS and submit screenshots of each step as evidence:

(i) Launch multiple EC2 instances (web servers) and configure them in different Availability Zones.

(ii) Create an Application Load Balancer (ALB) to distribute traffic across these instances.

(iii) Configure health checks so that faulty instances are automatically removed from load balancing.

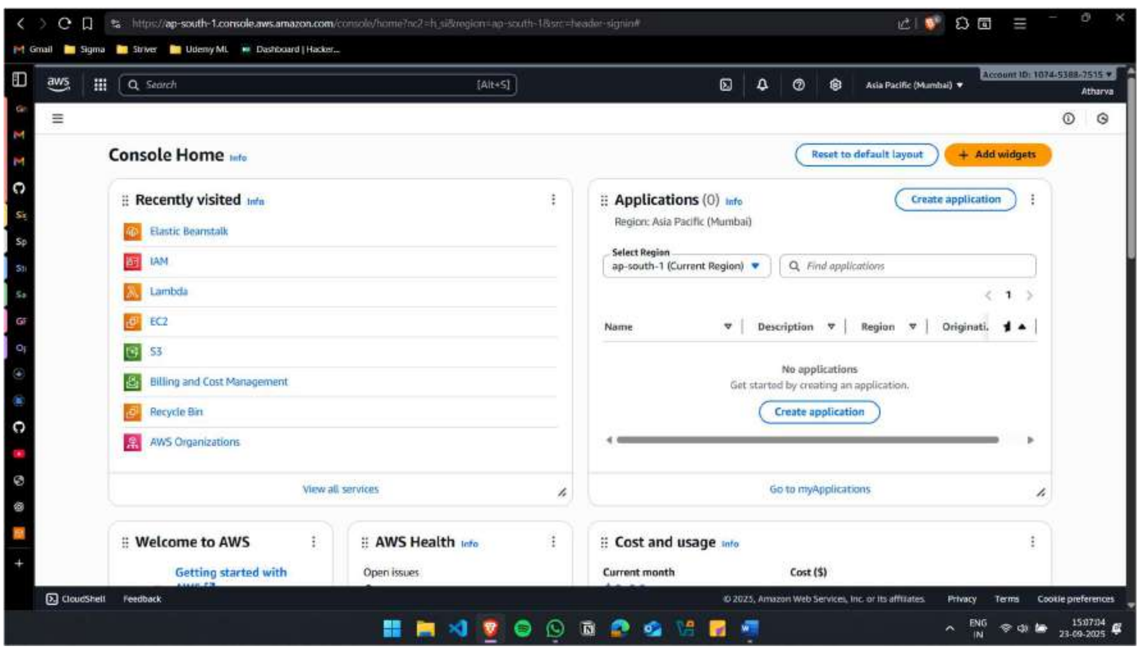
(iv) Enable Auto Scaling to add/remove instances based on traffic demand.

(v) Configure path-based routing: /auth requests go to the authentication service, /order requests go to the order processing service. Integrate the Load Balancer with Route 53 so that global users are routed to the nearest AWS region.

➔ **SOLUTION:**

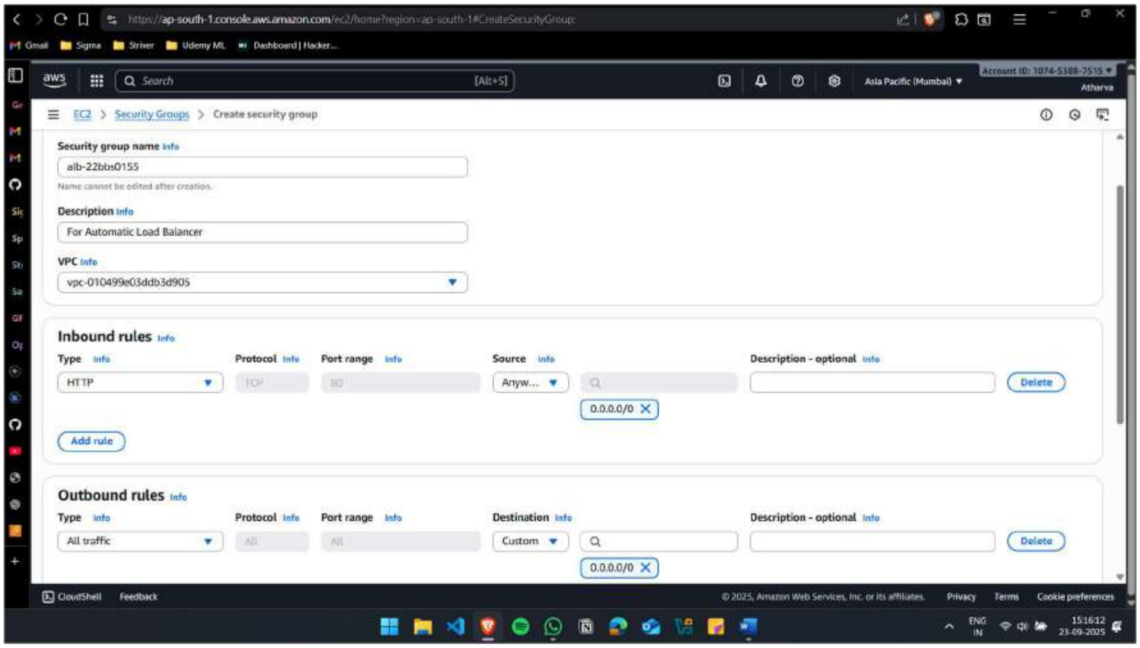
(i) Launch multiple EC2 instances (web servers) and configure them in different Availability Zones.

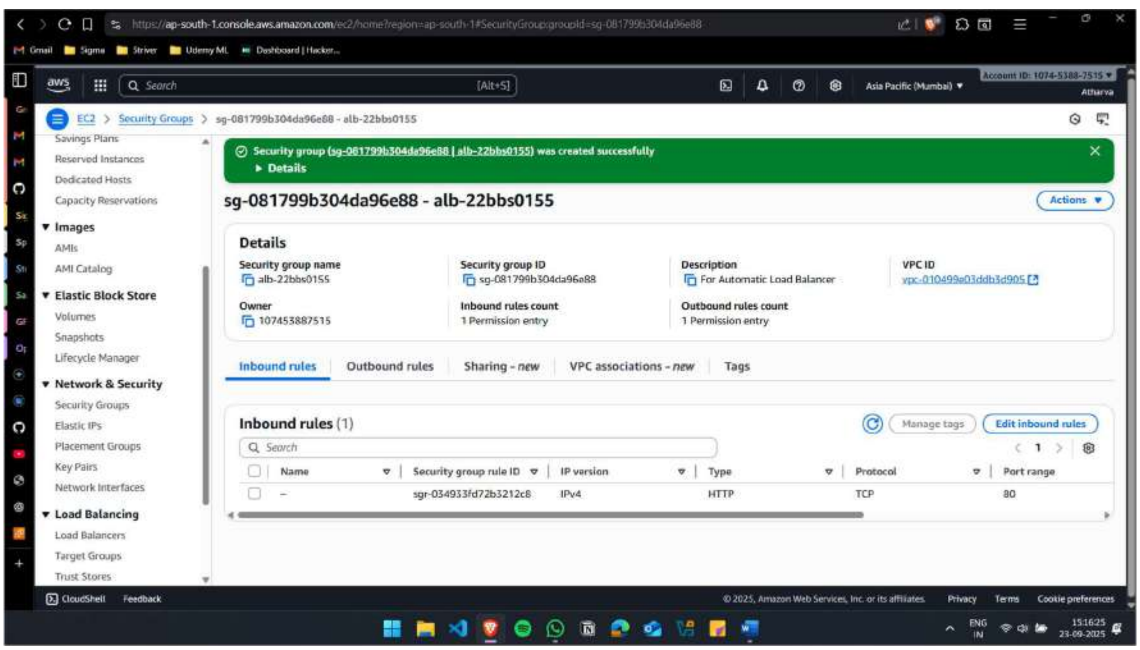
Initial:



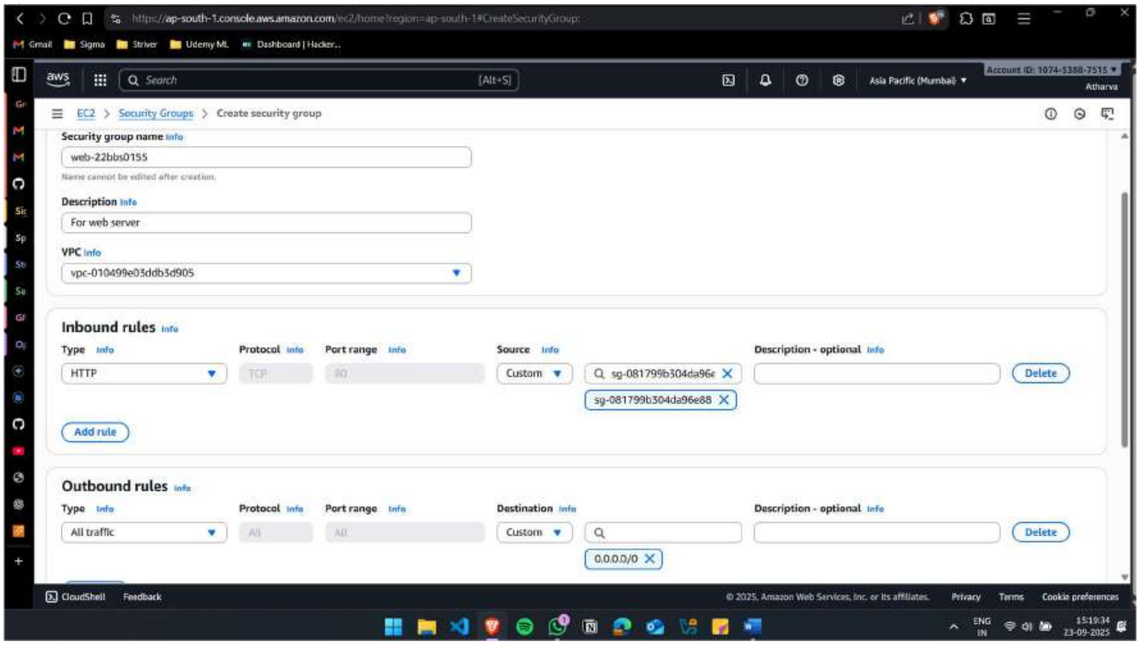
Create 2 security group for the instances:

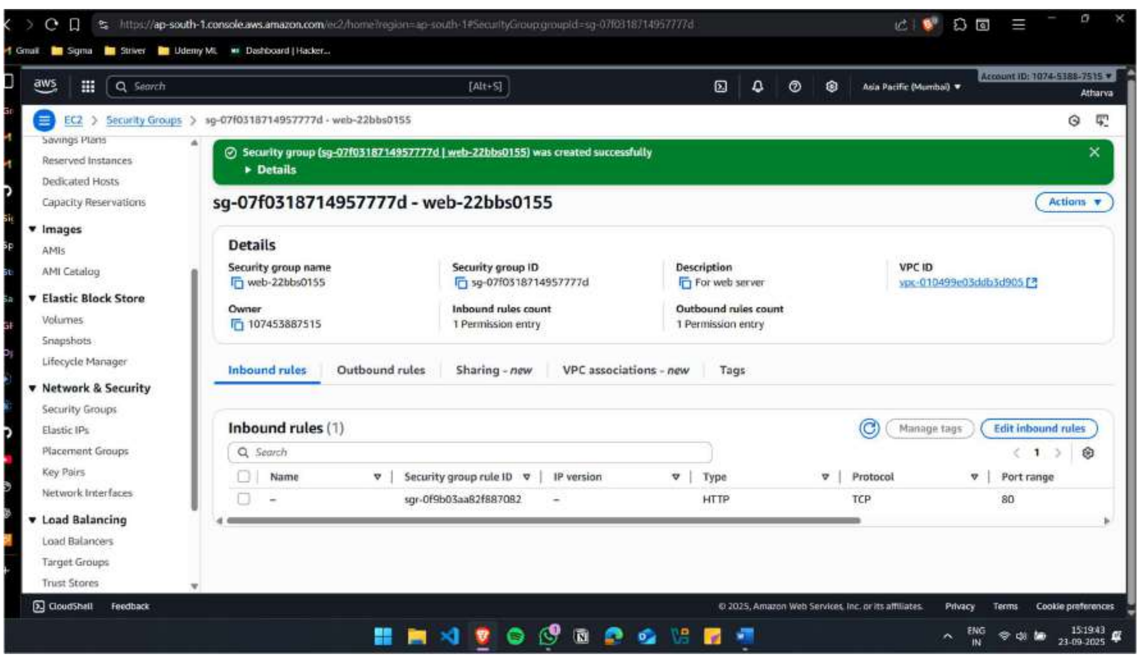
alb-22bbs0155: For Automatic Load Balancer





web-22bbs0155: For Web Server

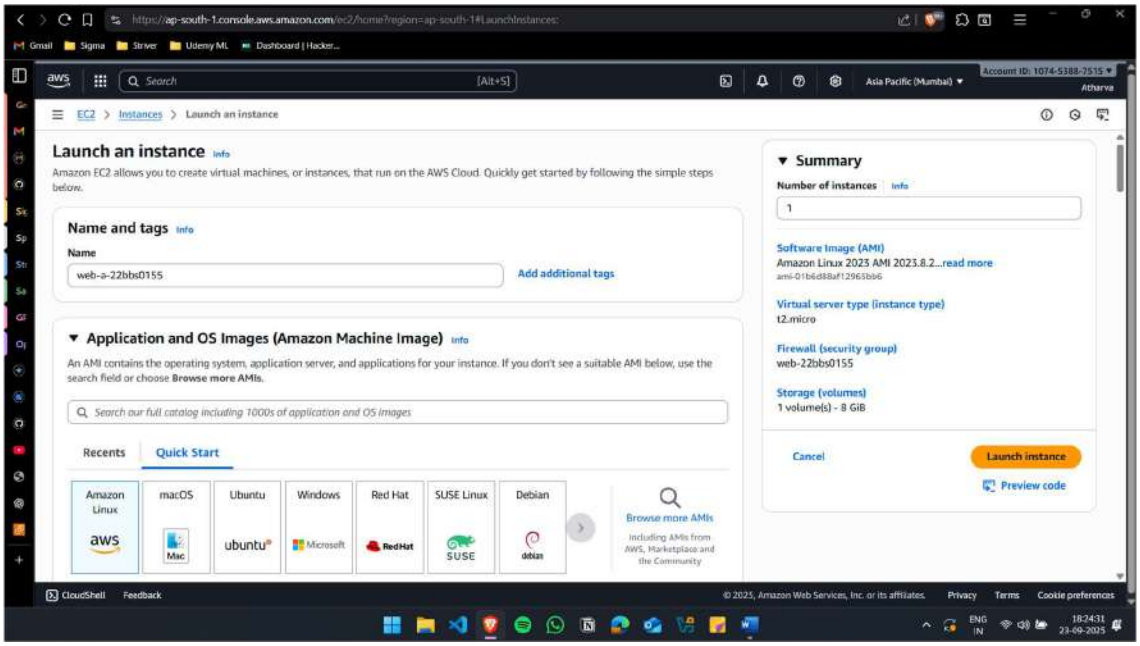


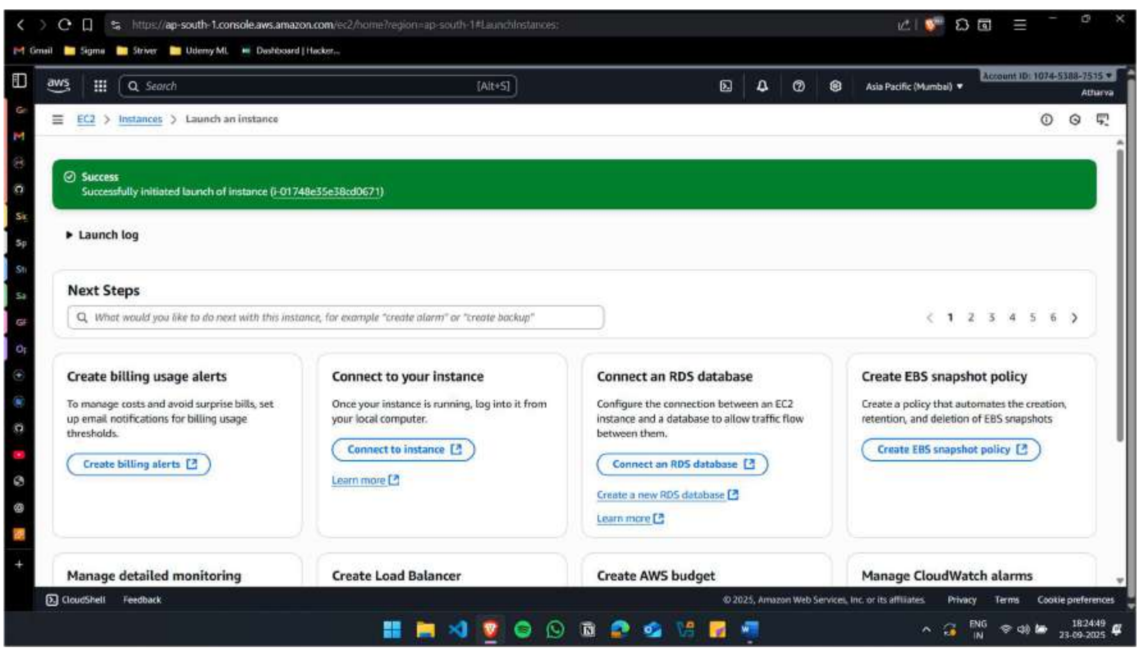


The web server can have inbound only from the sg-alb

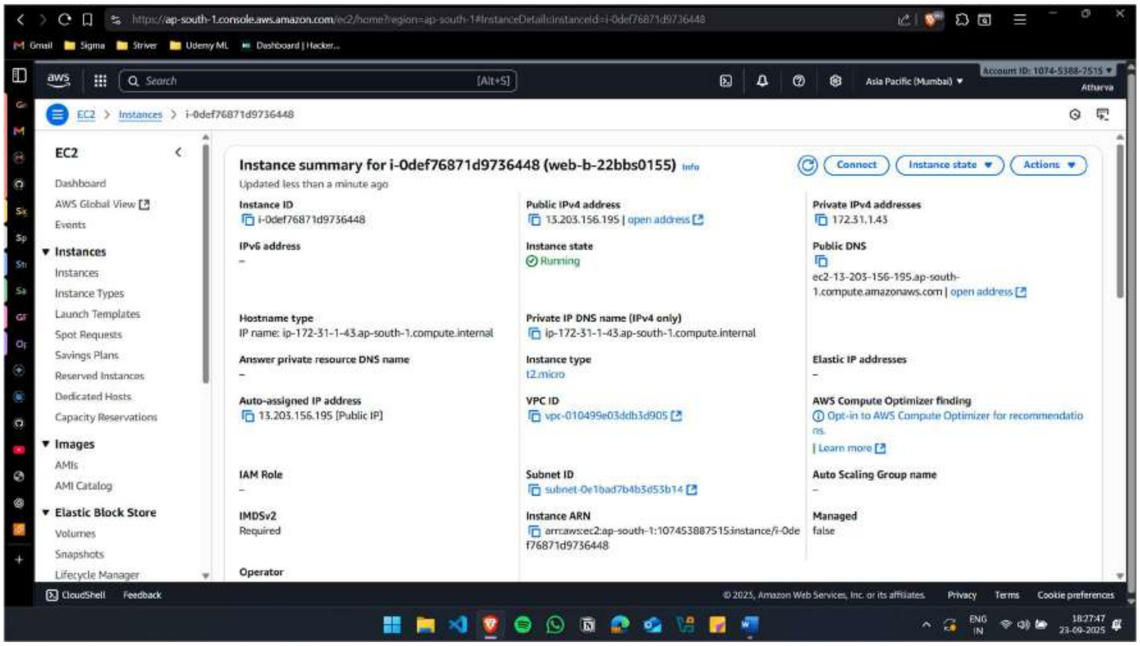
Create 2 EC2 Instances in different AZ

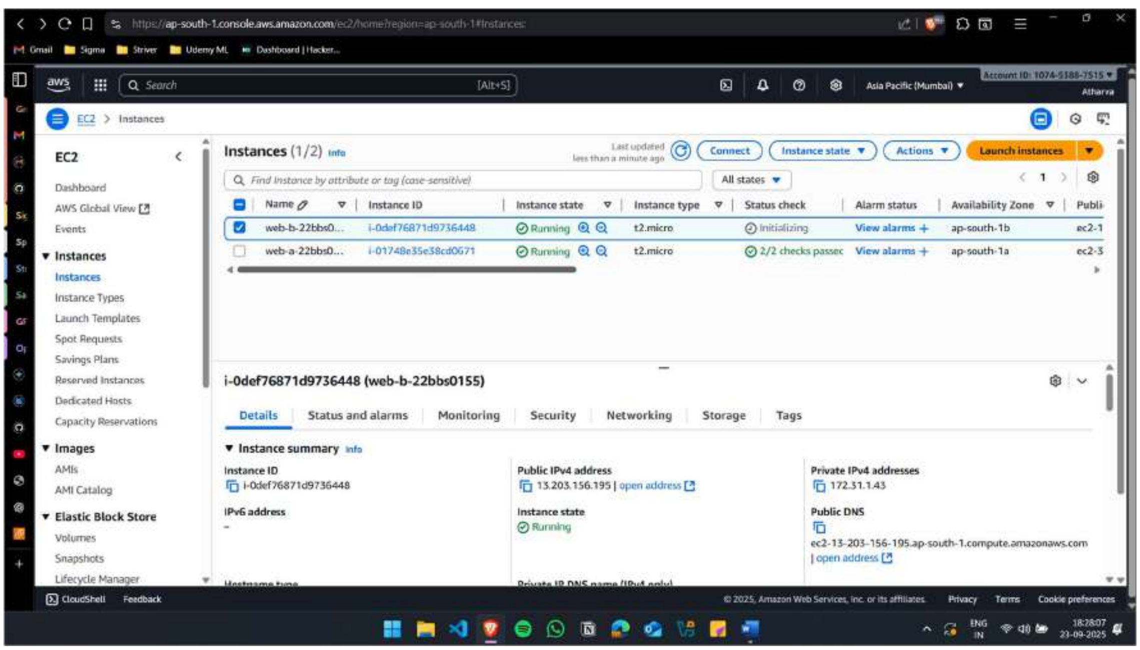
web-a-22bbs0155: AZ = ap-south-1a





web-b-22bbs0155: AZ = ap-south-1b

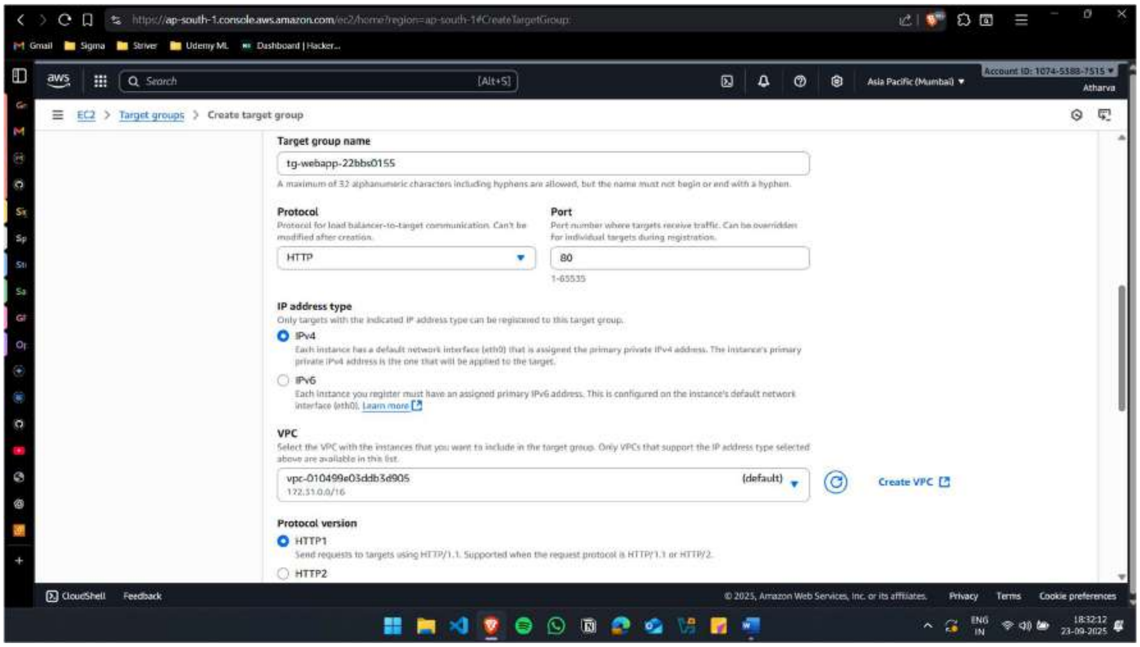


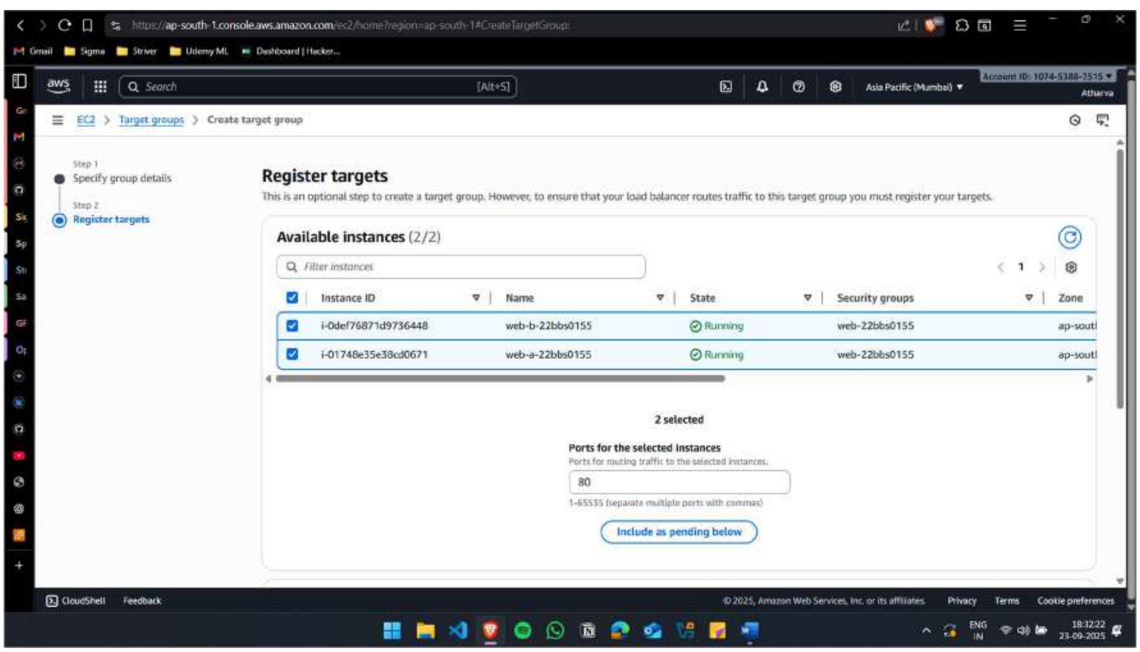


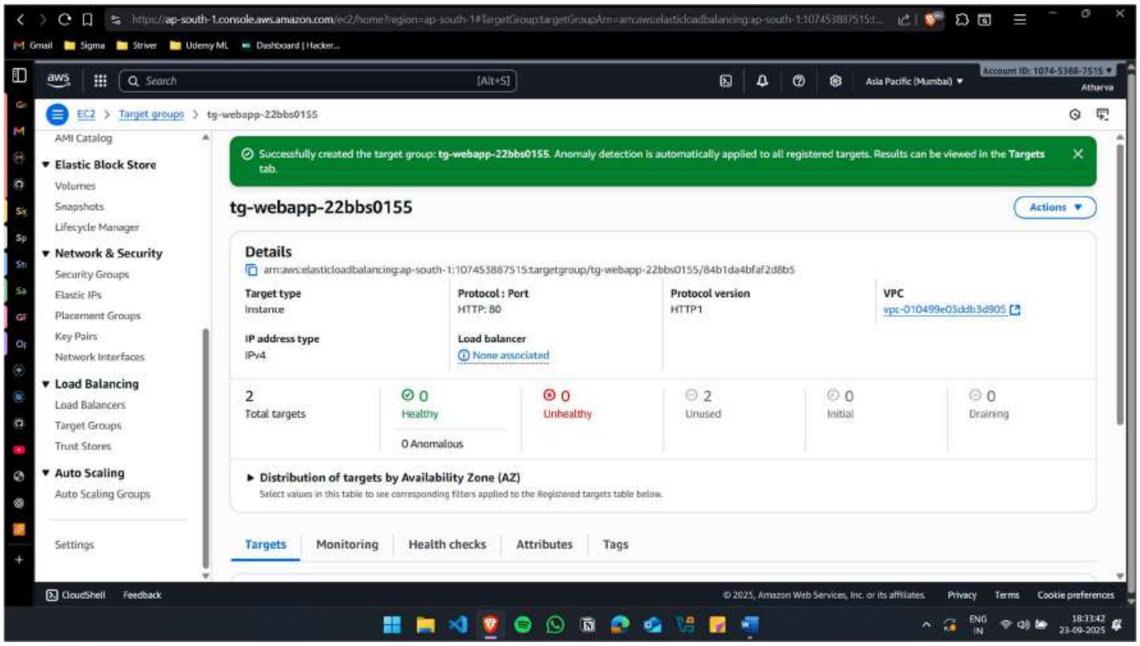
(ii) Create an Application Load Balancer (ALB) to distribute traffic across these instances.

Create Target Group

tg-webapp-22bbs0155

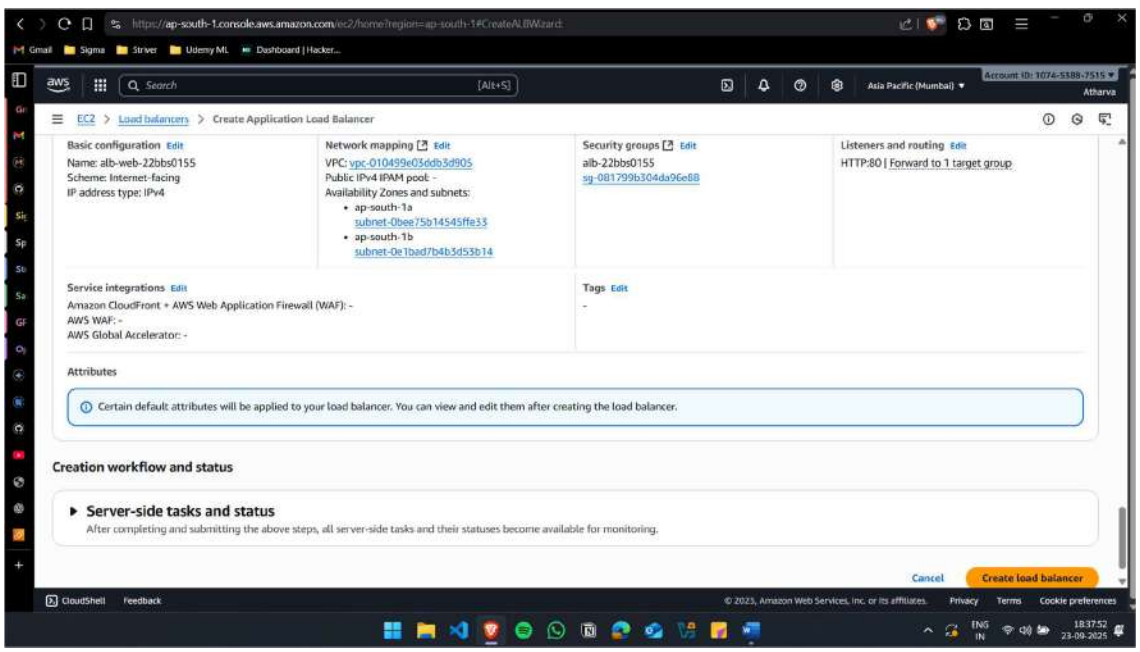


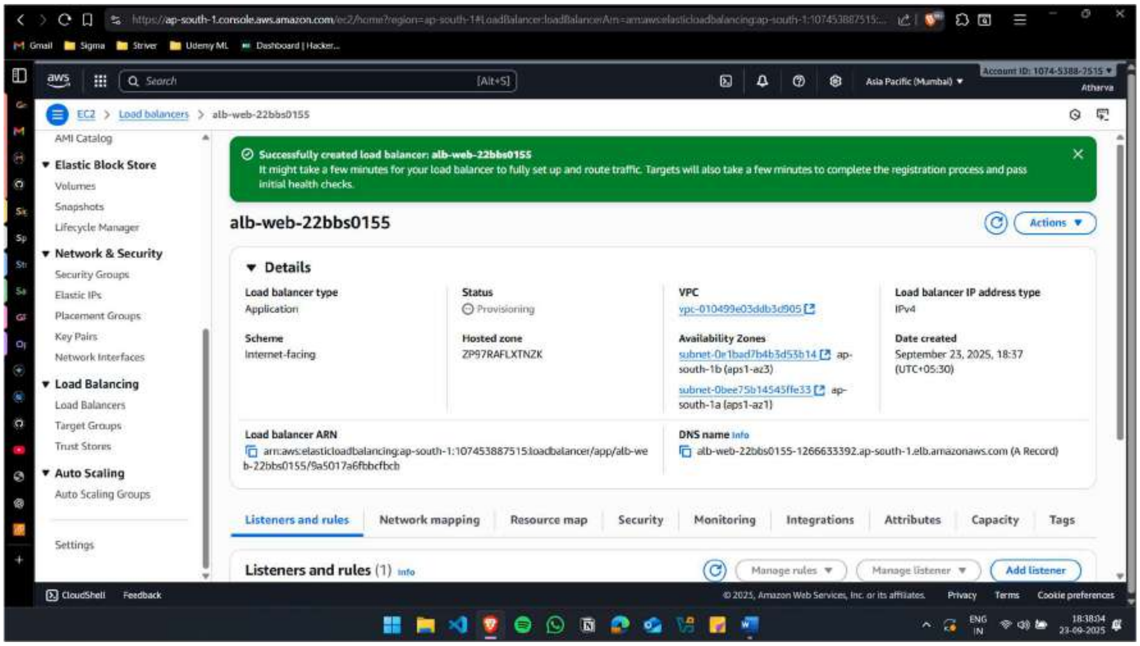




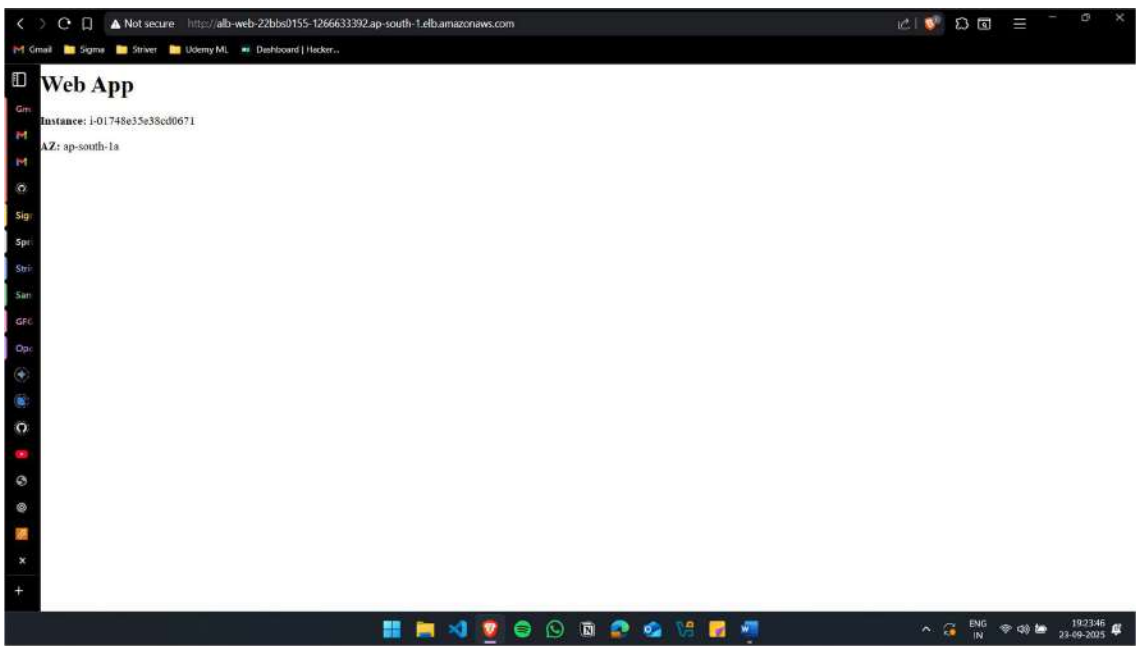
Create the Automatic Load Balancer

alb-web-22bbs0155





Accessing the Files from the ALB DNS (After refreshing)



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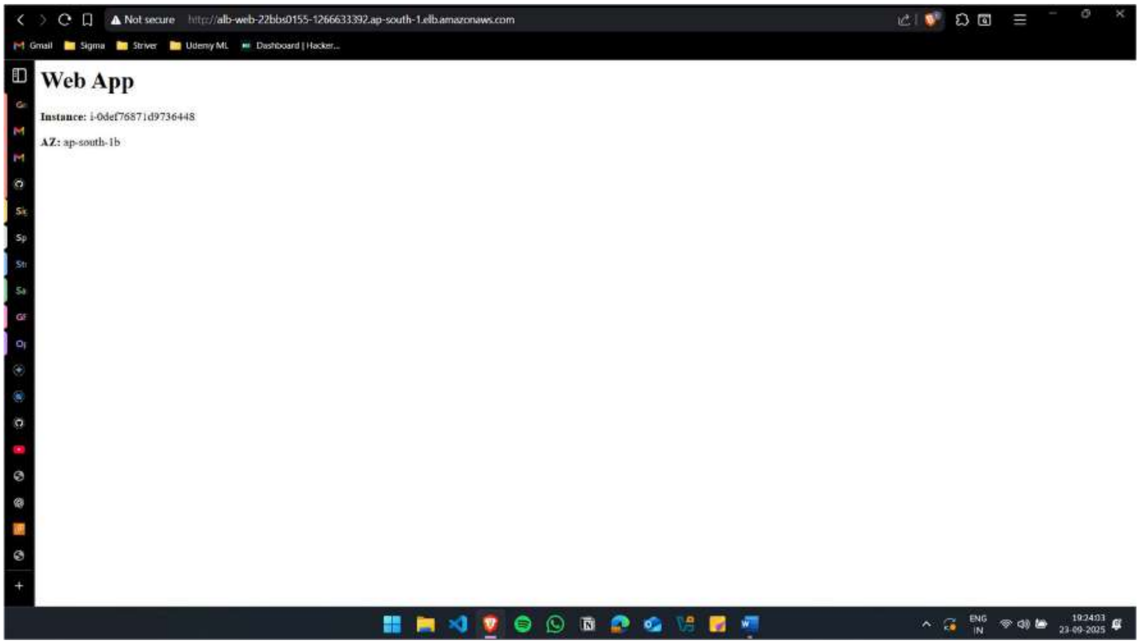
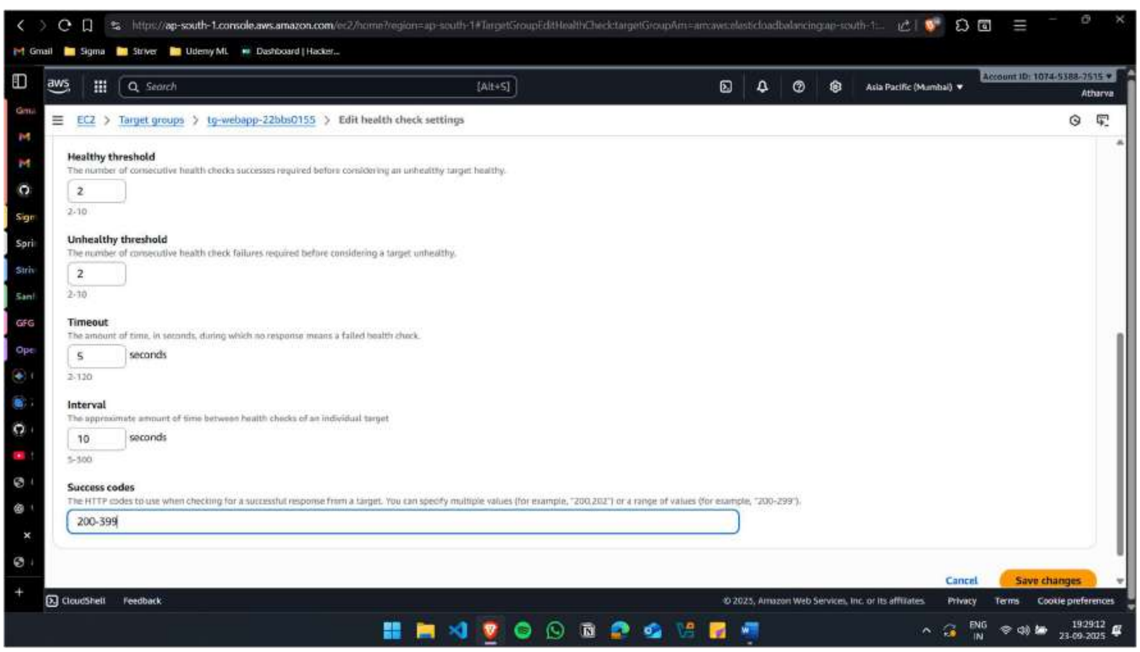
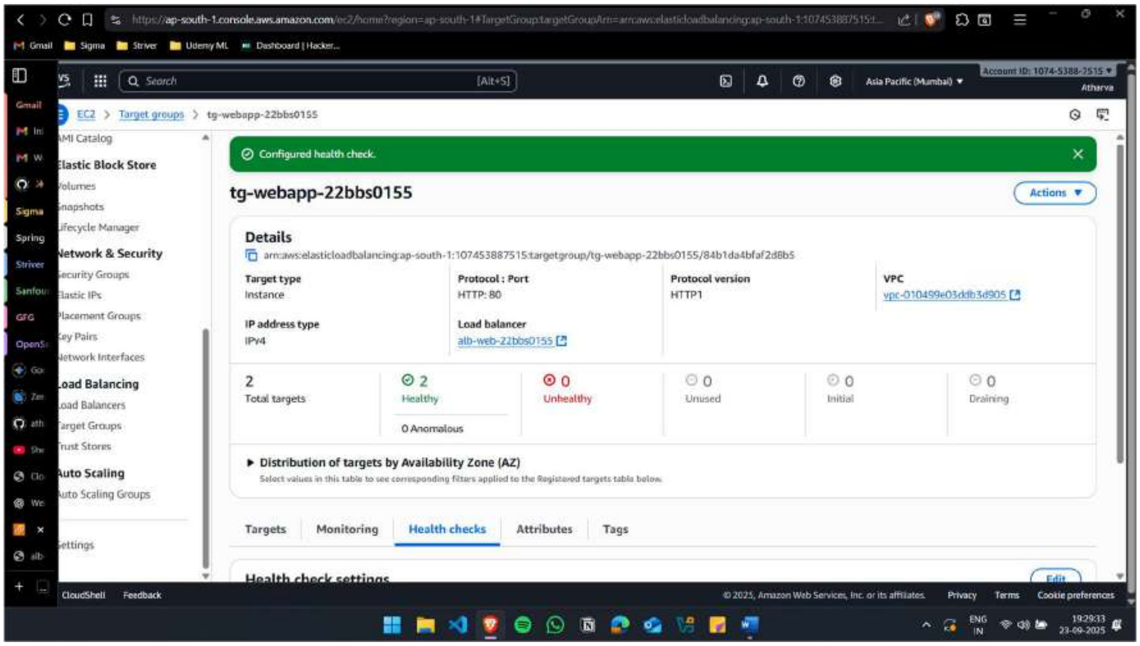


(iii) Configure health checks so that faulty instances are automatically removed from load balancing.

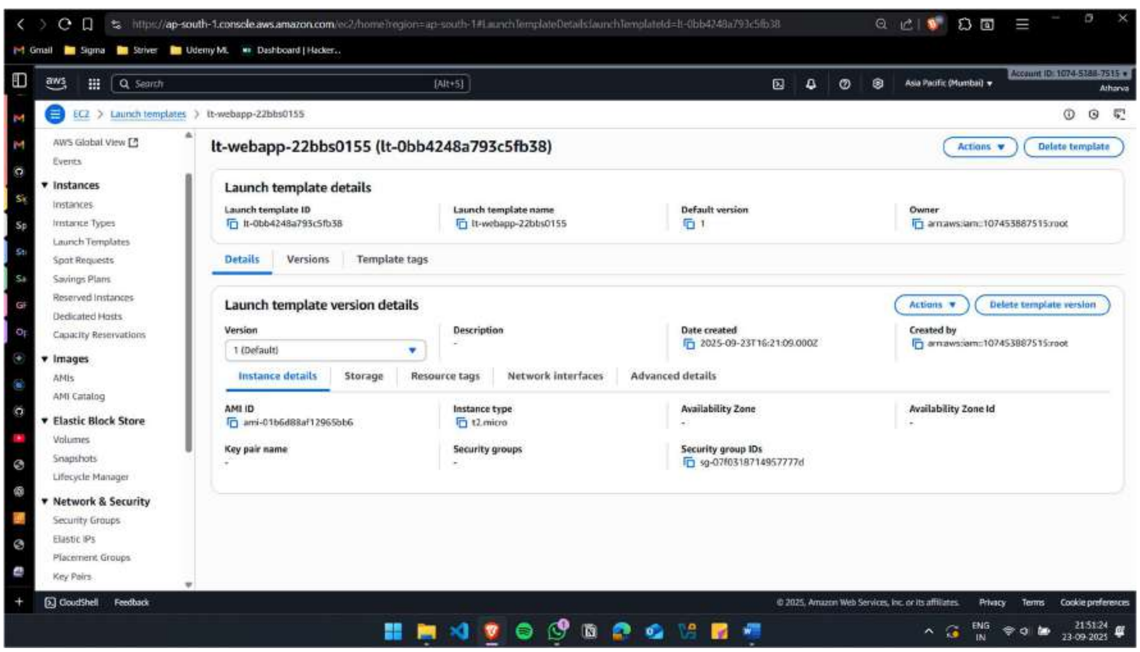
Edit Health Check Settings

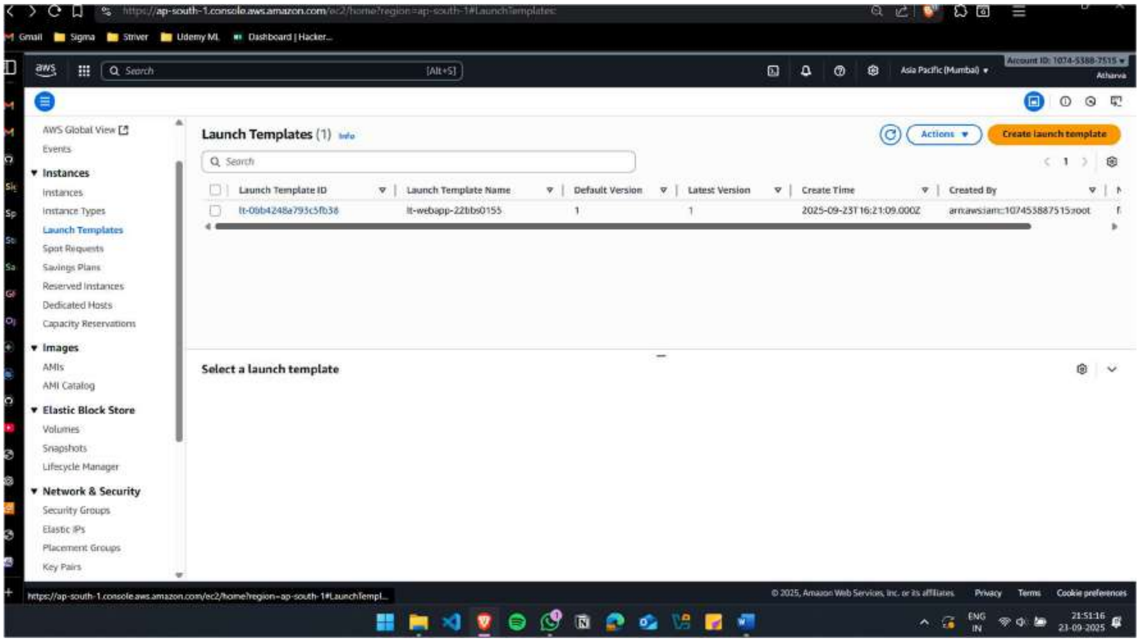




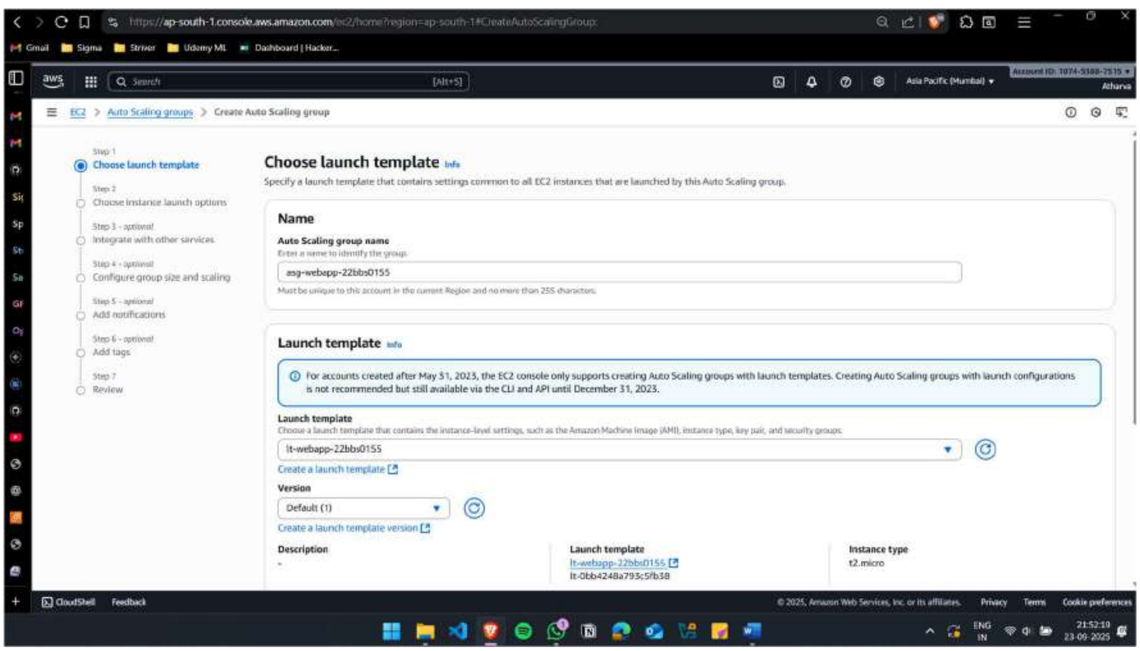
(iv) Enable Auto Scaling to add/remove instances based on traffic demand.

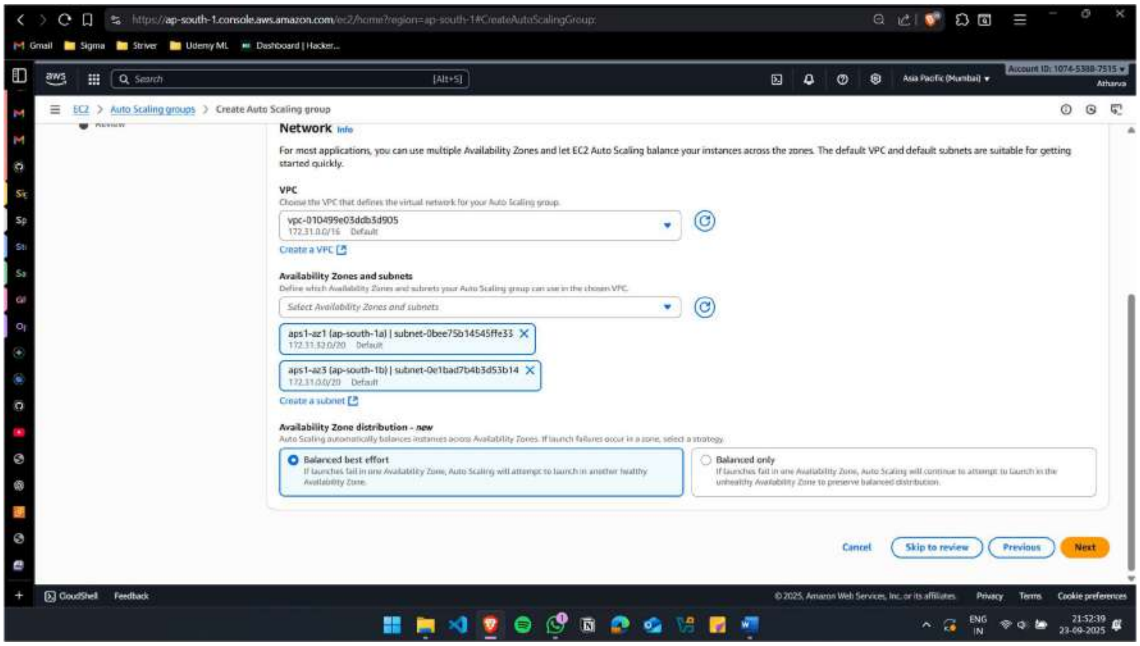
Create Launch Template

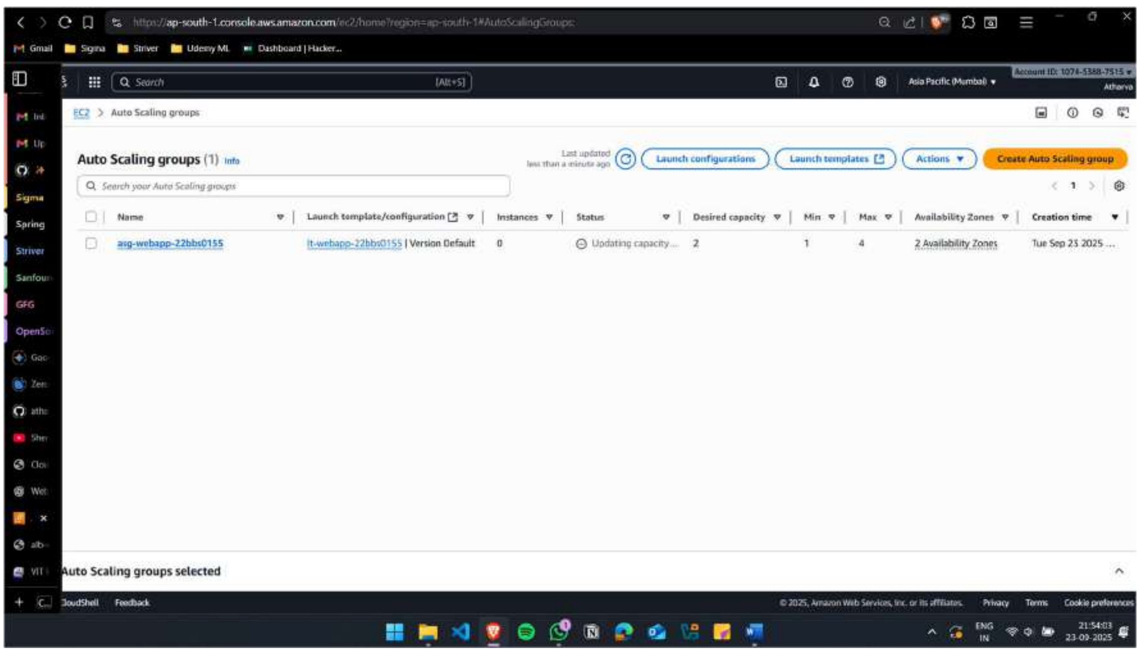


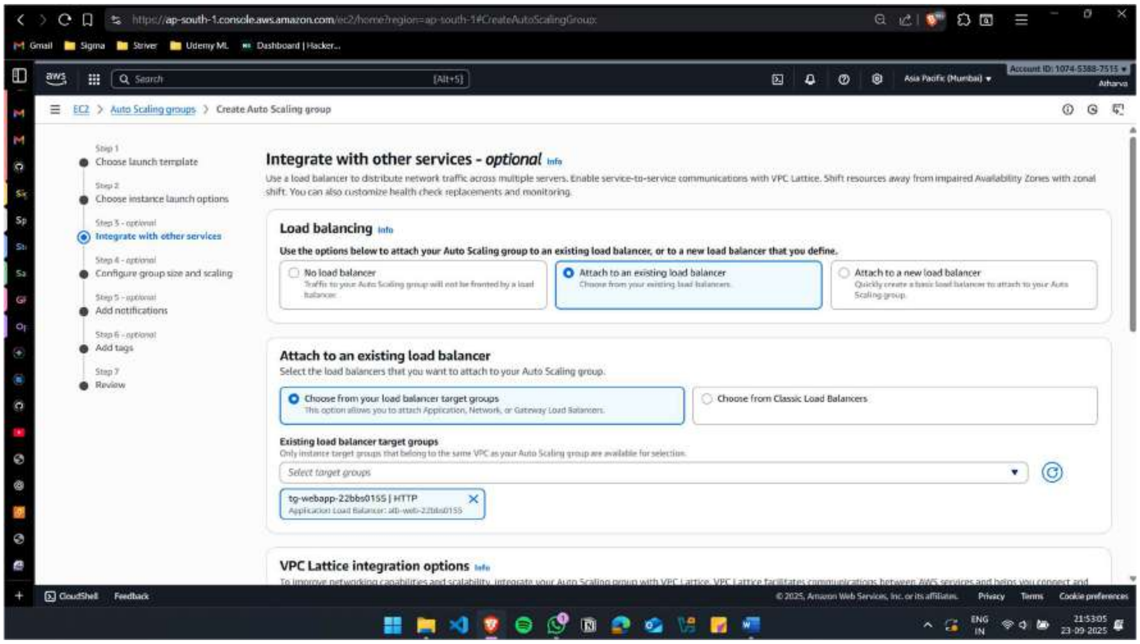


Create Auto Scaling Groups

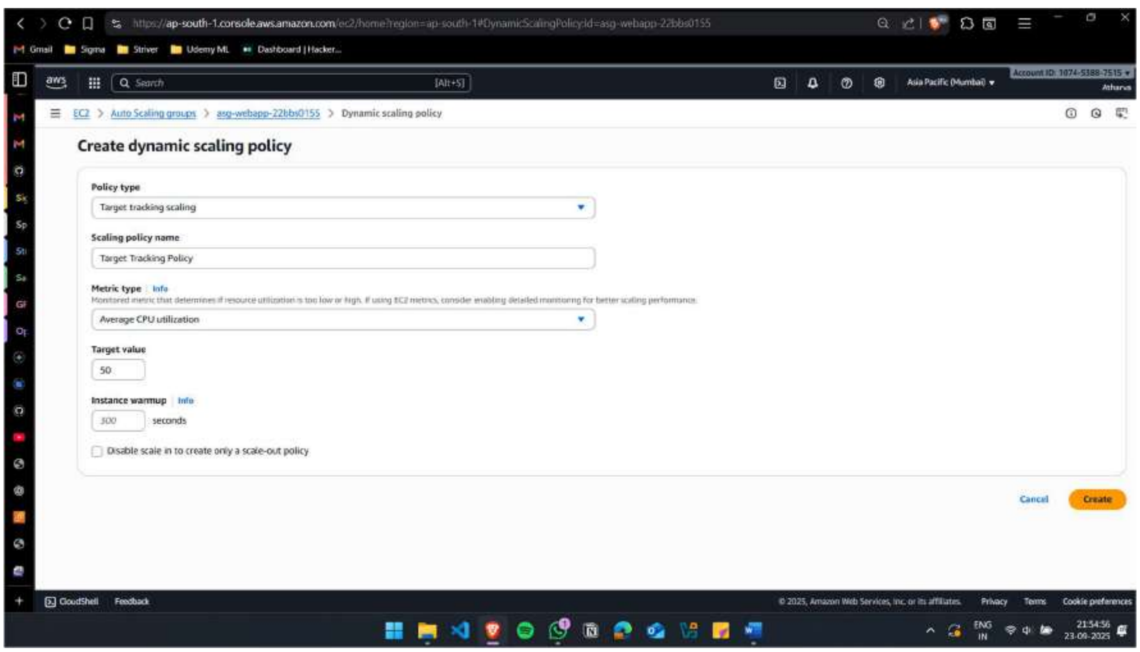


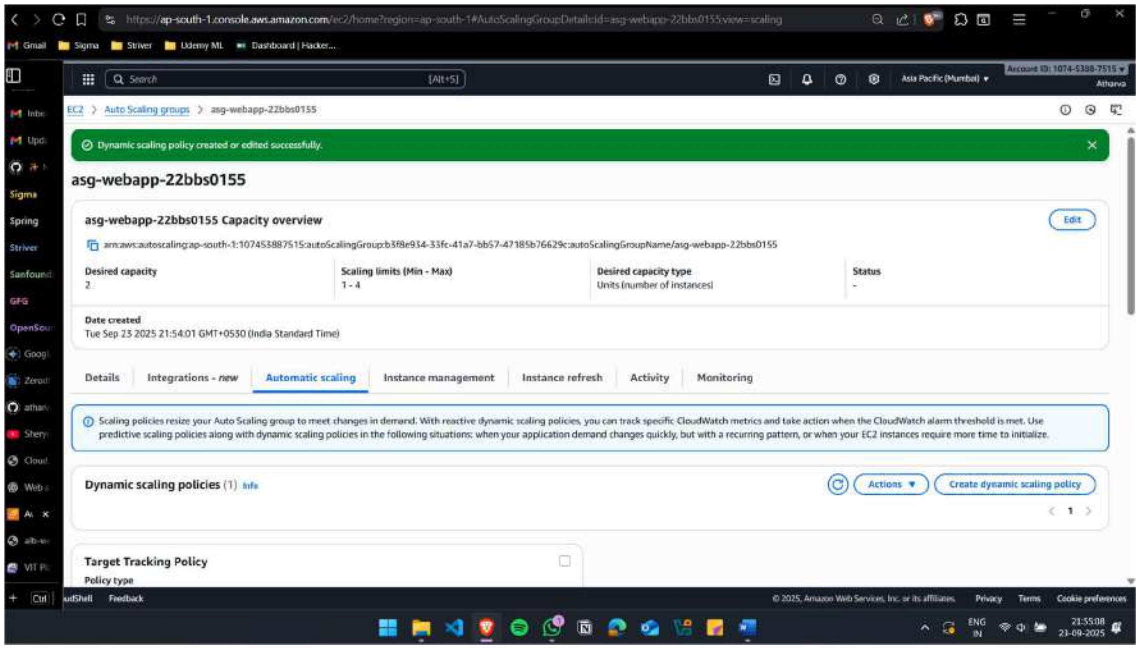

Network
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
vpc-010499e03ddb3d905
Availability Zones and subnets
aps1-az1 (ap-south-1a) | subnet-0bee75b14545ffe33
aps1-az3 (ap-south-1b) | subnet-0e1bad7b4b3d53b14
Availability Zone distribution - new
Balanced best effort
Balanced only
Cancel
Skip to review
Previous
Next


Auto Scaling groups (1)
Launch configurations
Launch templates
Actions
Create Auto Scaling group
Name
Launch template/configuration
Instances
Status
Desired capacity
Min
Max
Availability Zones
Creation time
asg-webapp-22bbs0155
lt-webapp-22bbs0155 | Version Default
0
Updating capacity...
2
1
4
2 Availability Zones
Tue Sep 23 2025 ...



Create Scaling Policy

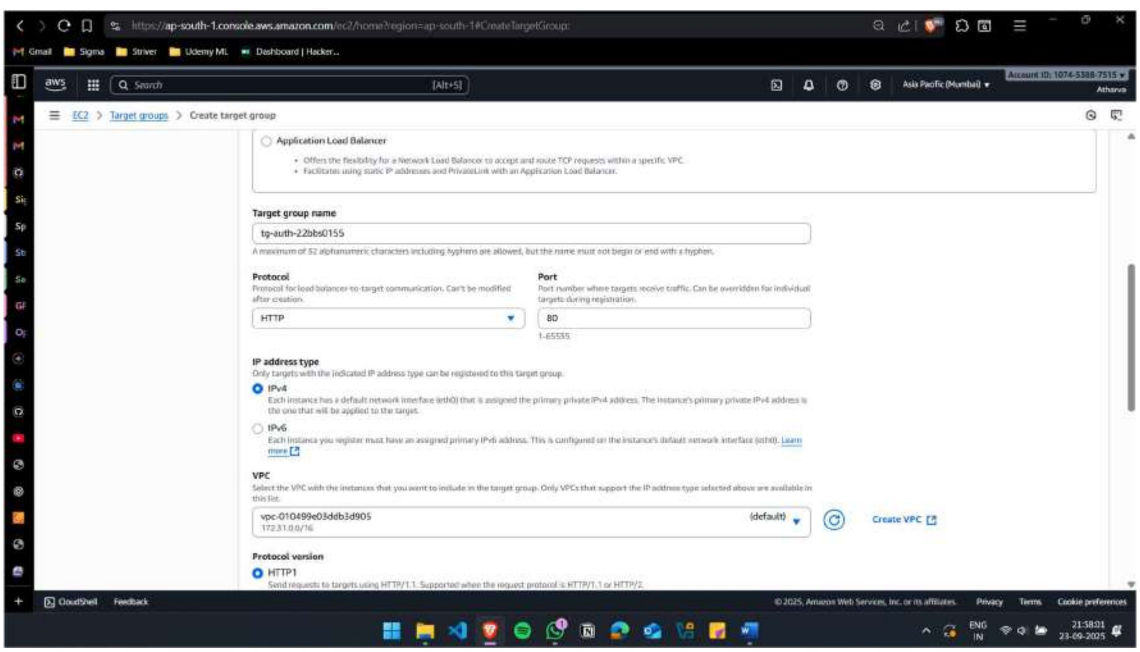


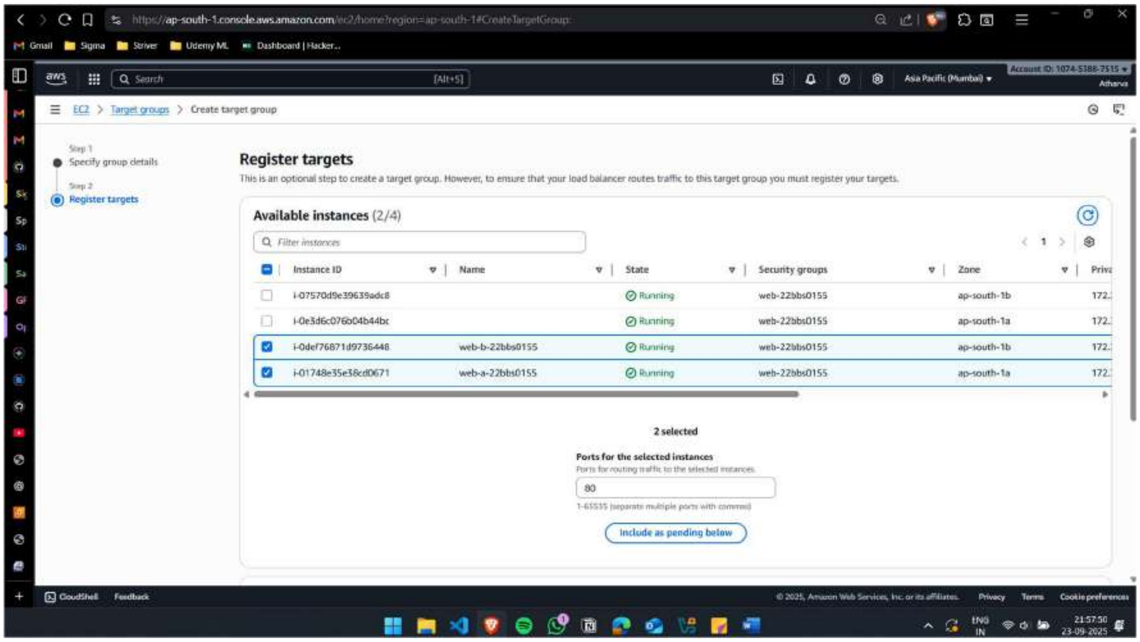


(v) Configure path-based routing: /auth requests go to the authentication service, /order requests go to the order processing service. Integrate the Load Balancer with Route 53 so that global users are routed to the nearest AWS region.

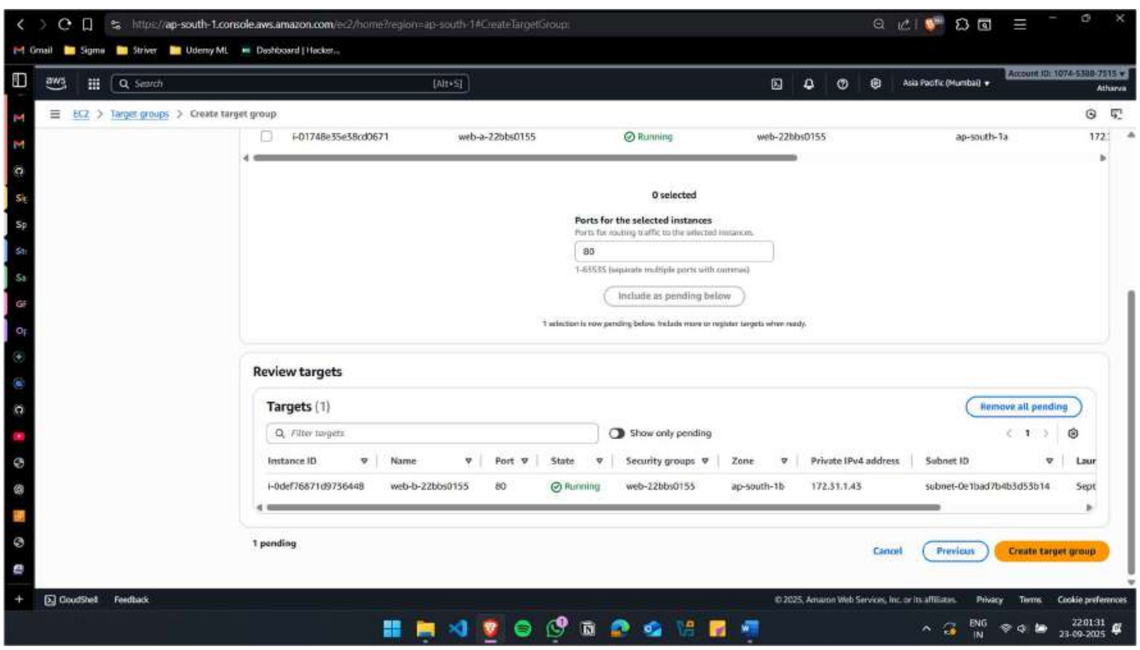
For this, instead of creating additional instances, I'm assuming web-a as Auth and web-2 as order.

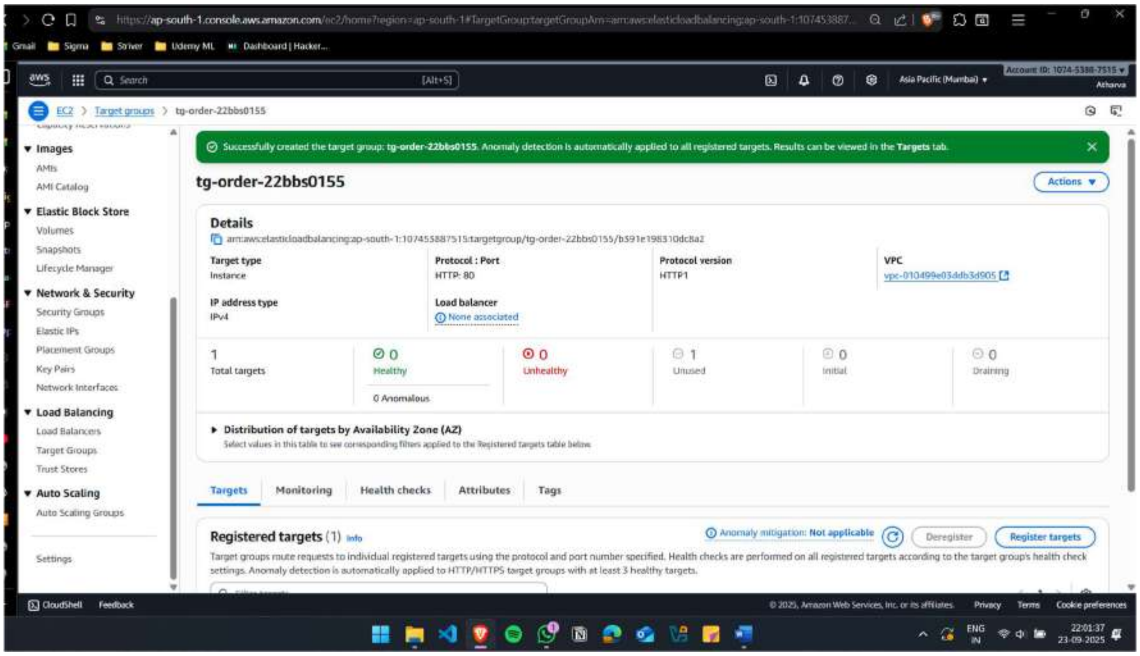
Create new Target Group for Auth(Web-a)



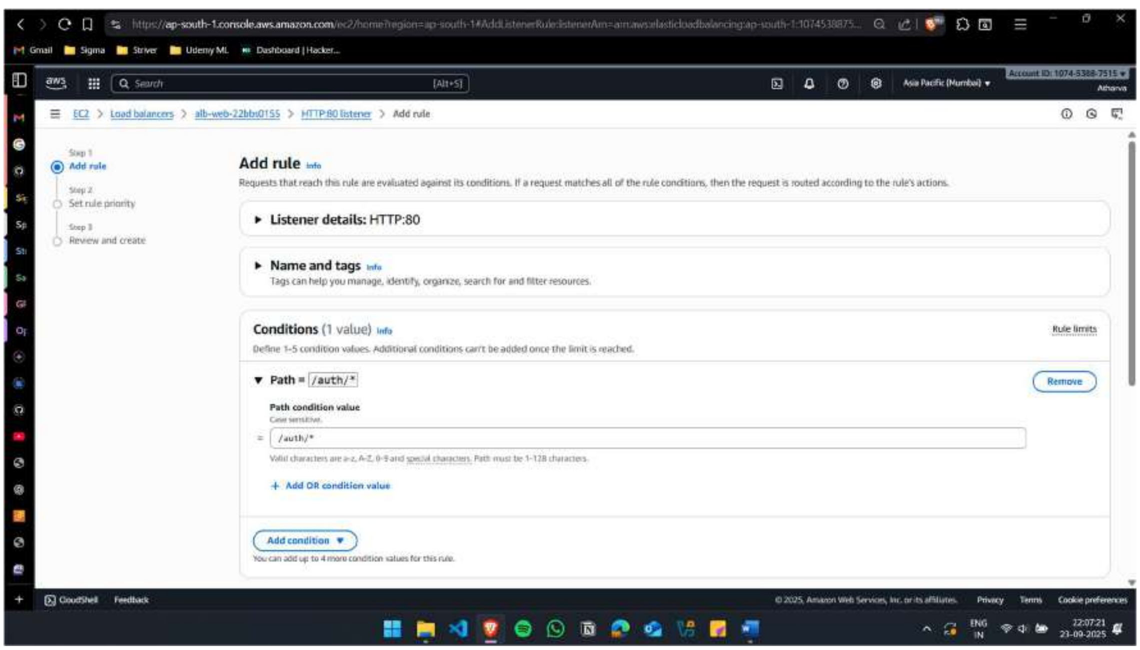


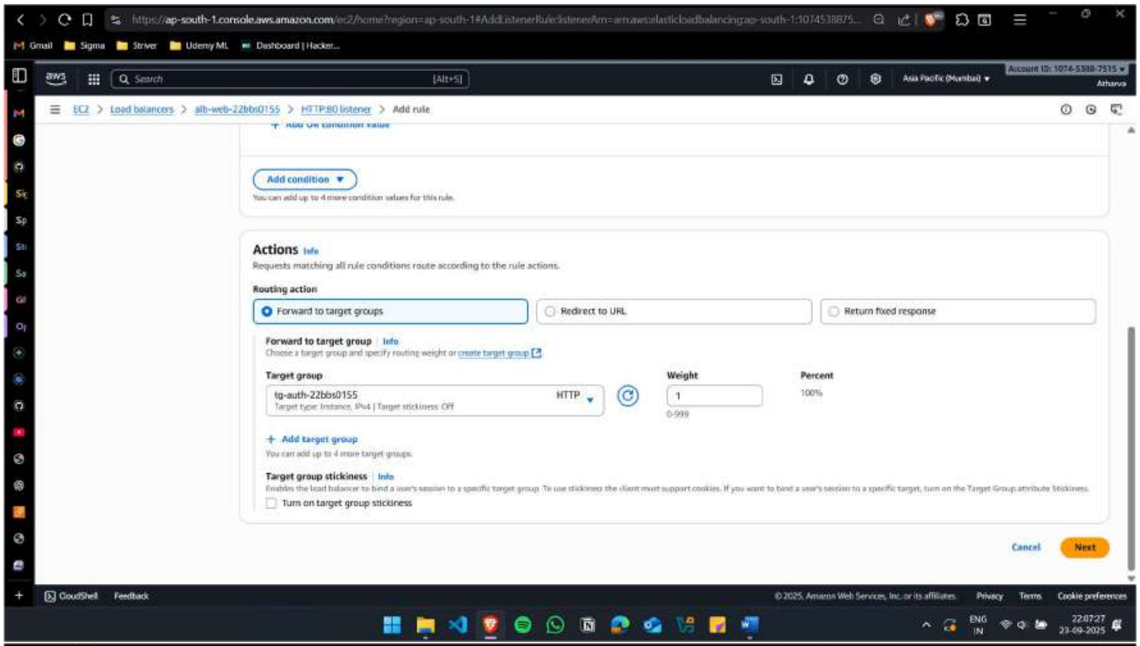
Create new Target Group for Order(Web-b)

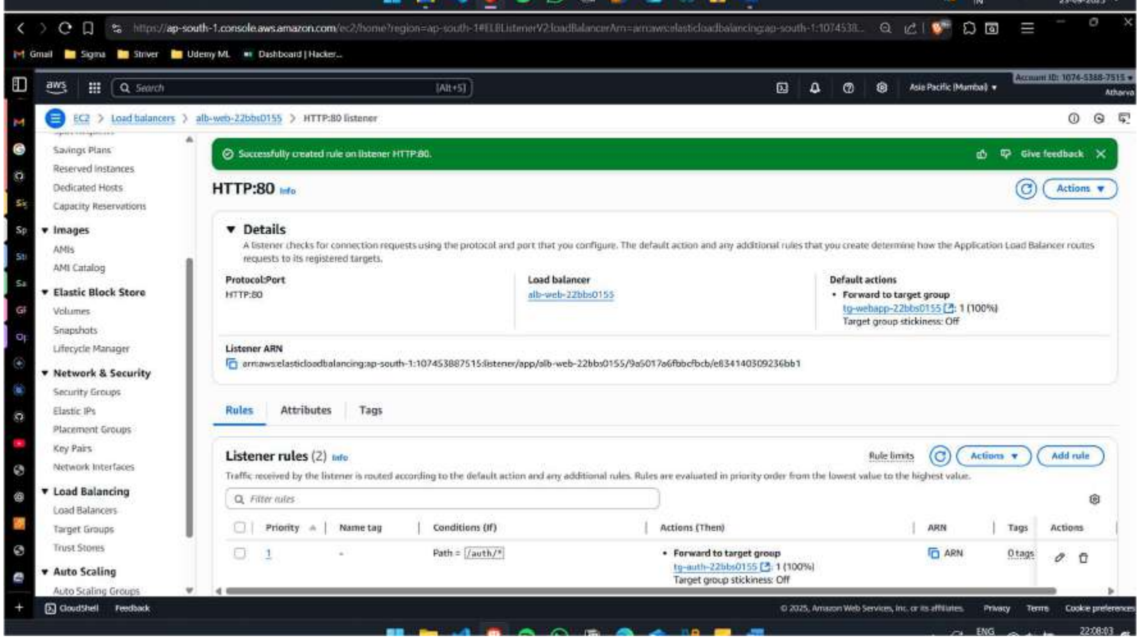


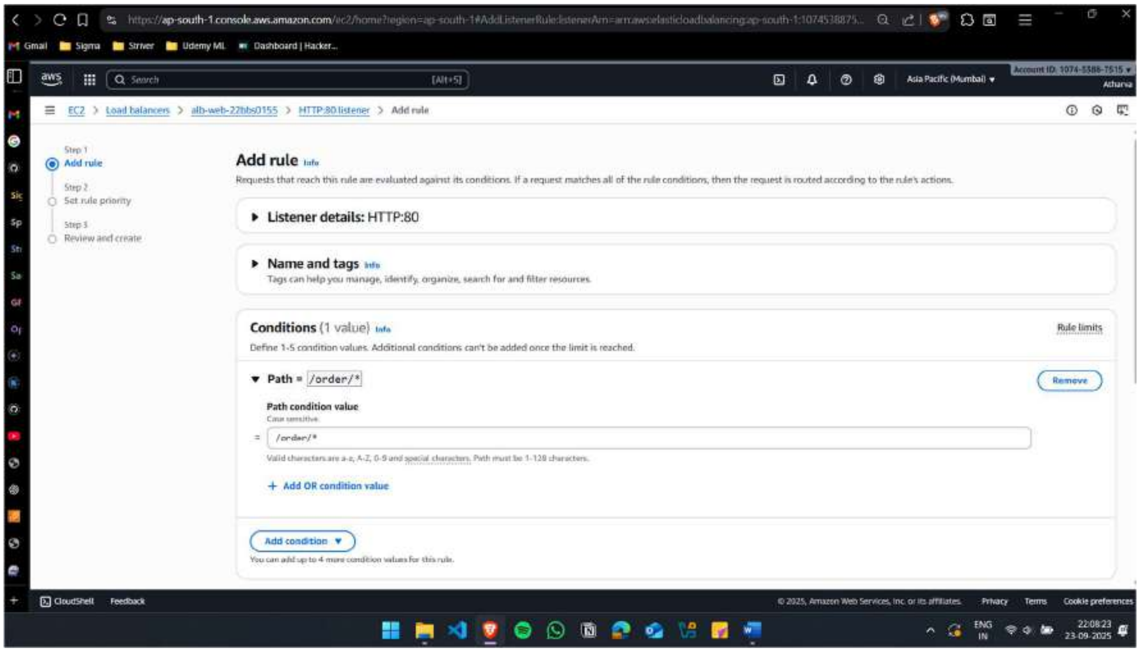


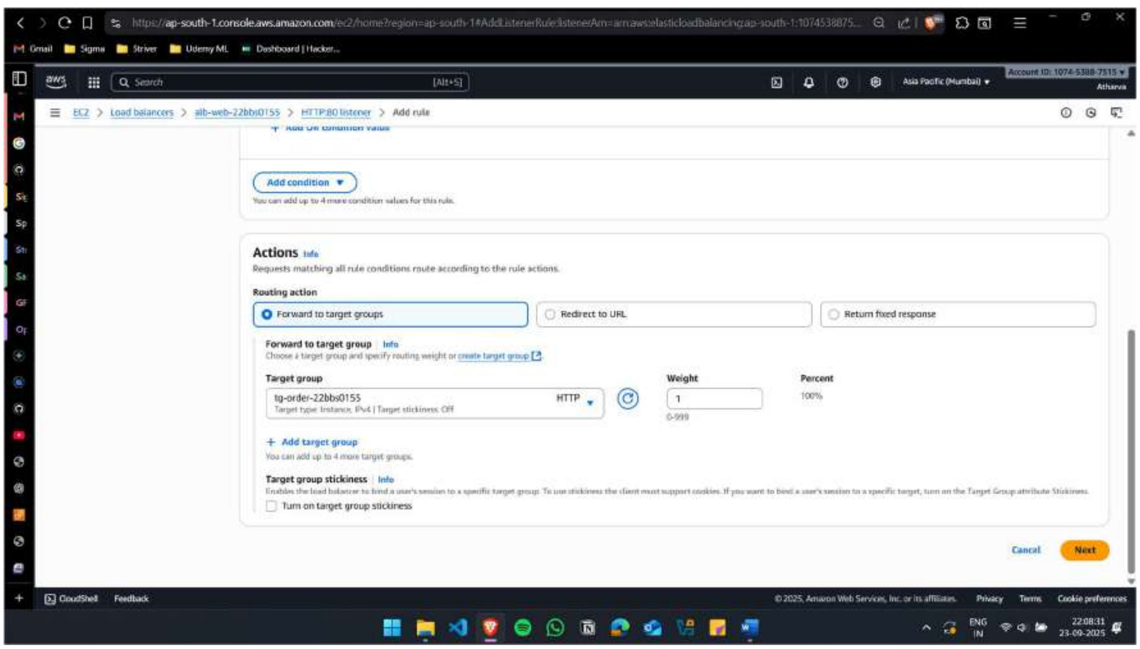
Add the rules to the ALB

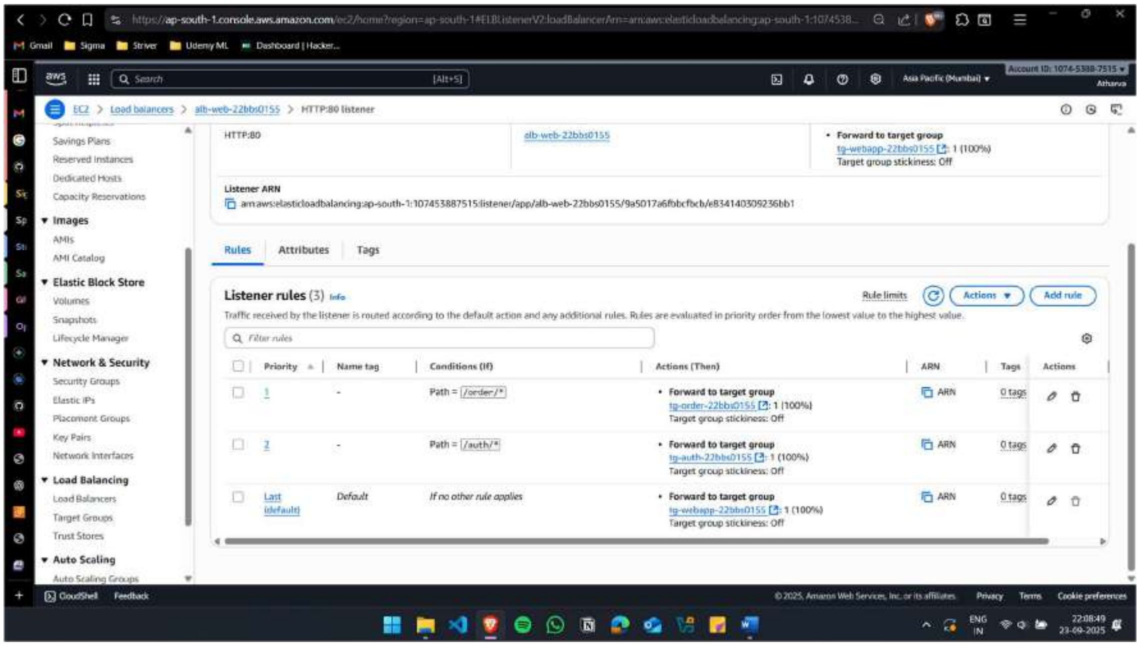



Add condition
You can add up to 4 more condition values for this rule.
Actions Info
Requests matching all rule conditions route according to the rule actions.
Routing action
Forward to target groups
Redirect to URL
Return fixed response
Forward to target group Info
Target group
tg-auth-22bbs0155
HTTP
Weight
1
Percent
100%
Add target group
Target group stickiness Info
Turn on target group stickiness
Cancel
Next

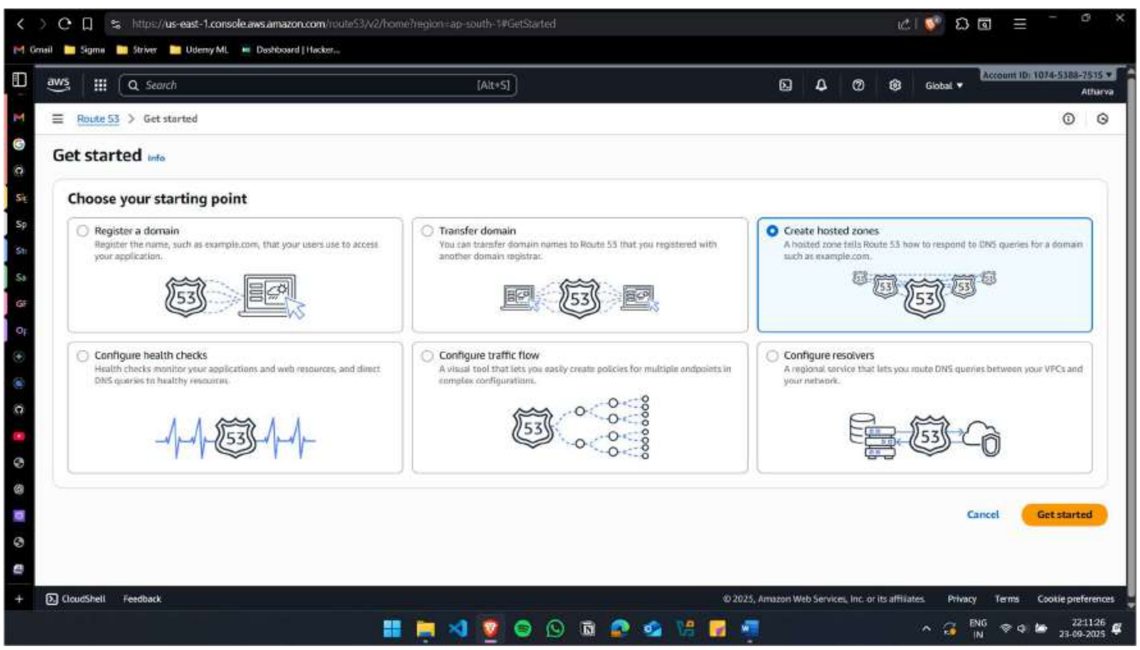

Successfully created rule on listener HTTP:80.
HTTP:80 Info
Details
Protocol:Port
HTTP:80
Load balancer
alb-web-22bbs0155
Default actions
Forward to target group
tg-webapp-22bbs0155: 1 (100%)
Target group stickiness: Off
Listener ARN
Rules
Attributes
Tags
Listener rules (2) Info
Priority
Name tag
Conditions (If)
Actions (Then)
1
Path = /auth/*
Forward to target group
tg-auth-22bbs0155: 1 (100%)
Target group stickiness: Off

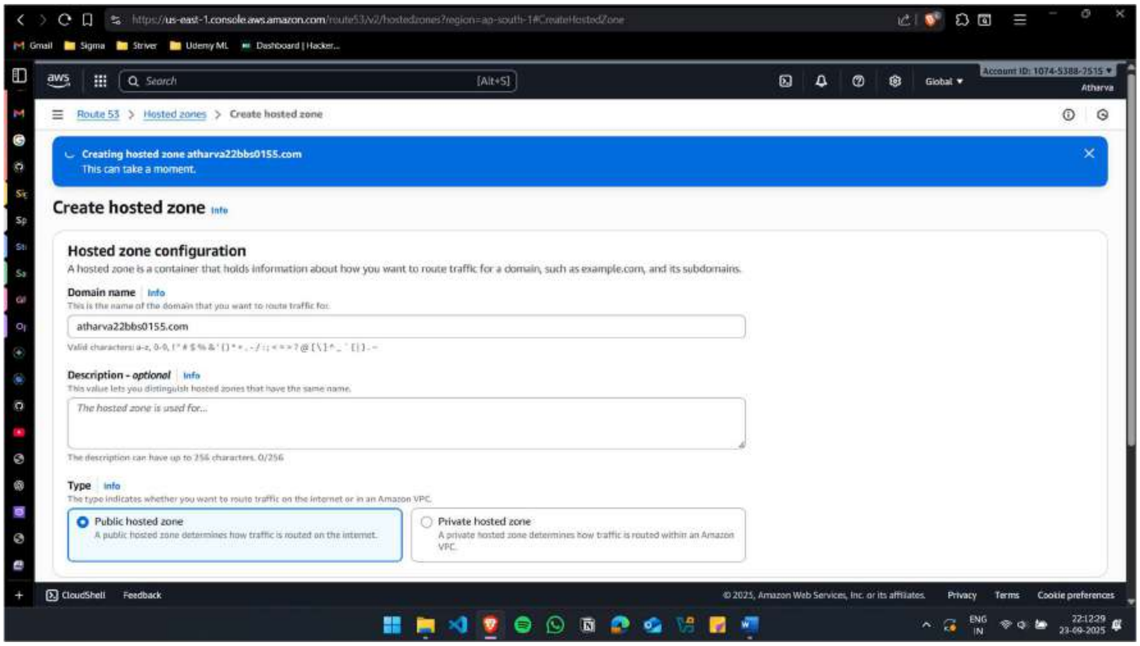

EC2 > Load balancers > alb-web-22bbs0155 > HTTP:80 listener > Add rule
Step 1
Add rule
Step 2
Set rule priority
Step 3
Review and create
Add rule Info
Requests that reach this rule are evaluated against its conditions. If a request matches all of the rule conditions, then the request is routed according to the rule's actions.
Listener details: HTTP:80
Name and tags Info
Tags can help you manage, identify, organize, search for and filter resources.
Conditions (1 value) Info
Rule limits
Define 1-5 condition values. Additional conditions can't be added once the limit is reached.
Path = /order/*
Remove
Path condition value
/order/*
+ Add OR condition value
Add condition
You can add up to 4 more condition values for this rule.
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


EC2 > Load balancers > alb-web-22bbs0155 > HTTP:80 listener > Add rule
Add condition
You can add up to 4 more condition values for this rule.
Actions Info
Requests matching all rule conditions route according to the rule actions.
Routing action
Forward to target groups
Redirect to URL
Return fixed response
Forward to target group Info
Target group
tg-order-22bbs0155
HTTP
Weight
1
Percent
100%
+ Add target group
You can add up to 4 more target groups.
Target group stickiness Info
Turn on target group stickiness
Cancel
Next
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

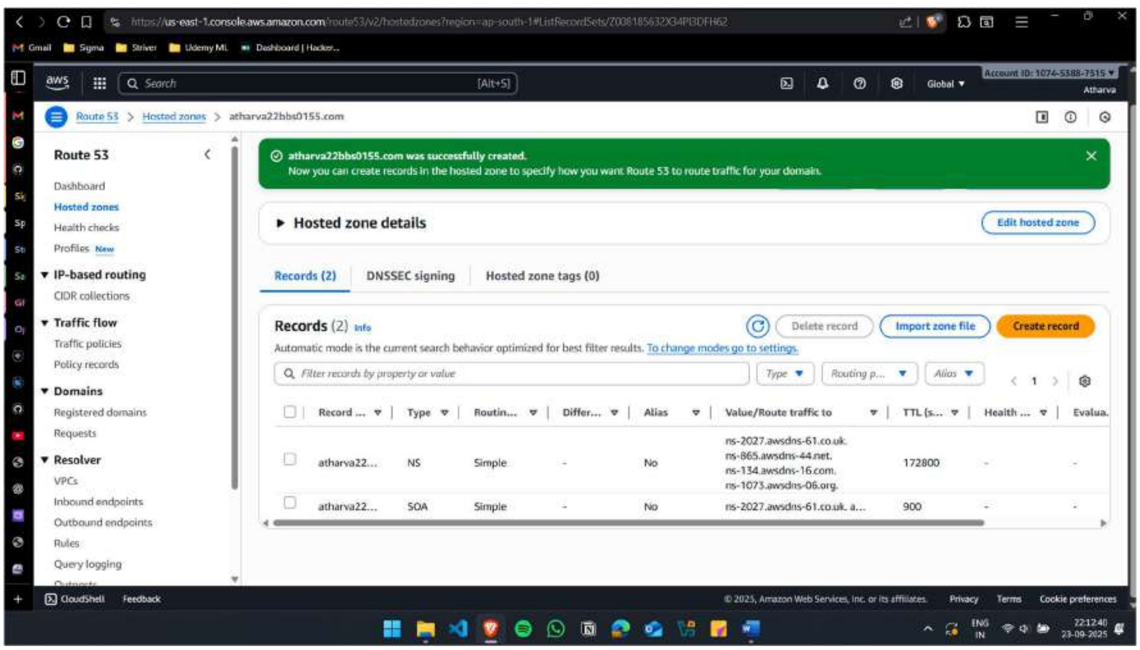


Route to the nearest hosted region



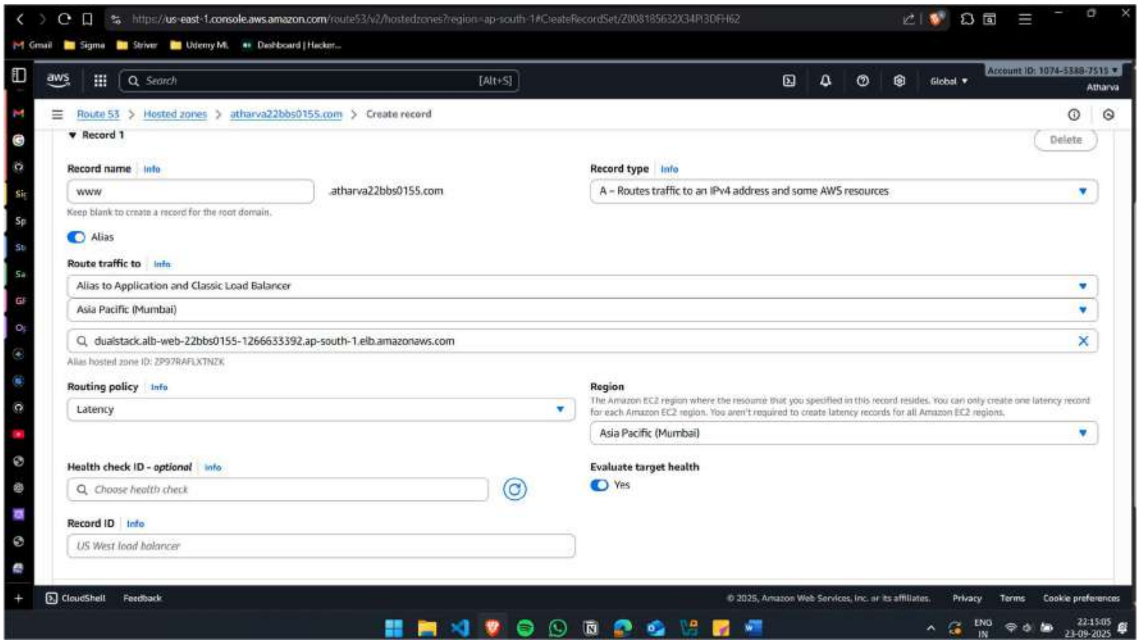


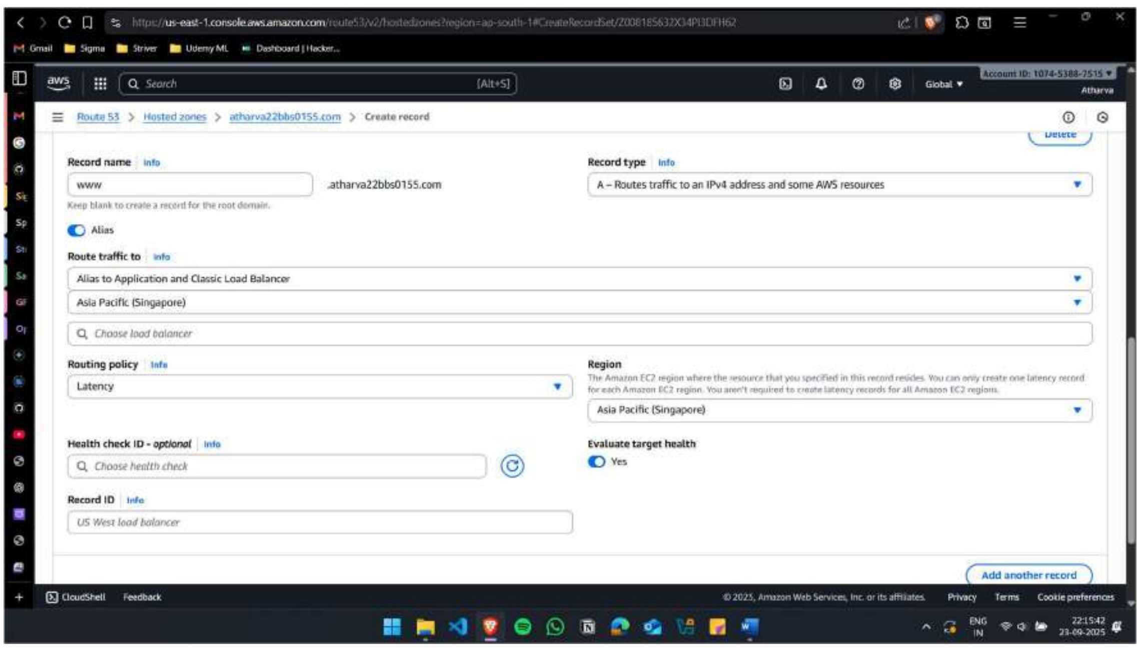
Creating hosted zone atharva22bbs0155.com
This can take a moment.
Create hosted zone
Hosted zone configuration
A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.
Domain name
atharva22bbs0155.com
Description - optional
Type
Public hosted zone
Private hosted zone

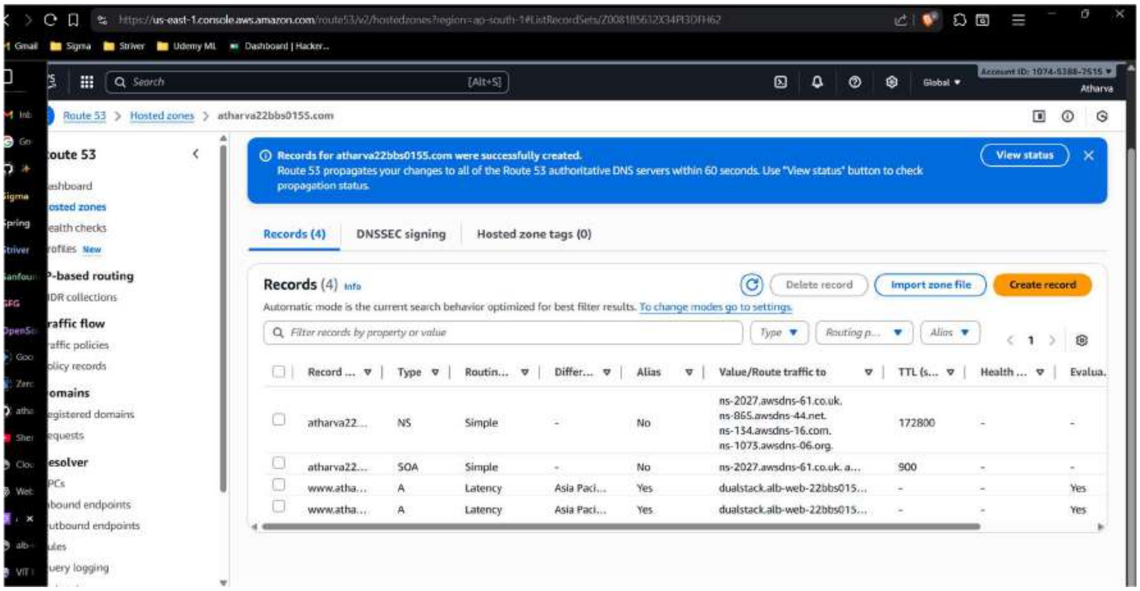


atharva22bbs0155.com was successfully created.
Now you can create records in the hosted zone to specify how you want Route 53 to route traffic for your domain.
Hosted zone details
Edit hosted zone
Records (2)
DNSSEC signing
Hosted zone tags (0)
Delete record
Import zone file
Create record
atharva22... NS Simple - No ns-2027.awsdns-61.co.uk. ns-865.awsdns-44.net. ns-134.awsdns-16.com. ns-1073.awsdns-06.org. 172800
atharva22... SOA Simple - No ns-2027.awsdns-61.co.uk. a... 900

Create 2 records(mumbai and singapore)






Route 53 > Hosted zones > atharva22bbs0155.com
Records for atharva22bbs0155.com were successfully created.
Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.
View status
Records (4)
DNSSEC signing
Hosted zone tags (0)
Records (4)
Delete record
Import zone file
Create record
Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.
Filter records by property or value
Record ...
Type
Routin...
Differ...
Alias
Value/Route traffic to
TTL (s...
Health ...
Evalua...
atharva22...
NS
Simple
No
ns-2027.awsdns-61.co.uk.
ns-865.awsdns-44.net.
ns-134.awsdns-16.com.
ns-1073.awsdns-06.org.
172800
atharva22...
SOA
Simple
No
ns-2027.awsdns-61.co.uk. a...
900
www.atha...
A
Latency
Asia Paci...
Yes
dualstack.alb-web-22bbs015...
Yes
Dashboard
Hosted zones
Health checks
IP-based routing
Traffic flow
Domains
Resolver
Inbound endpoints
Outbound endpoints
Query logging